

Hospital Management Analytics Dashboard – Project Proposal

1. Executive Summary

This project aims to analyze and visualize hospital operational data using Power BI to improve healthcare management and decision-making. The dataset includes information related to patients, doctors, appointments, treatments, and billing records.

The main goal of this project is to create an interactive Hospital Management Analytics Dashboard that helps hospitals monitor patient flow, doctor performance, appointment trends, treatment costs, and revenue generation.

The dashboard will provide real-time insights that can support hospital administrators in improving operational efficiency, reducing delays, and enhancing patient satisfaction.

2. Problem Statement

Background: Hospitals manage a large amount of patient and operational data daily. Without proper analytics, it becomes difficult to track appointments, monitor billing, analyze doctor performance, and manage hospital resources efficiently.

Objective: To develop an interactive Power BI dashboard that analyzes hospital management data and provides meaningful insights into hospital operations, patient records, appointments, treatments, and revenue performance.

Scope: The project focuses on patient analysis, doctor performance tracking, appointment management, treatment and billing analysis, and revenue monitoring.

3. Data Sources

Primary Data: N/A

Secondary Data: The dataset used for this project is the Hospital Management Dataset obtained from Kaggle. The dataset contains patient records, doctor information, appointments, treatments, billing, and payment details.

Source Link:

<https://www.kaggle.com/datasets/kanakbaghel/hospital-management-dataset>

4. Methodology

Data Collection: Importing the dataset from Kaggle into Power BI.

Data Preparation: Handling missing values, removing duplicates, correcting data types, and creating relationships between tables.

Data Modeling: Creating relationships between patients, doctors, appointments, treatments, and billing tables.

Analysis Techniques: Descriptive analytics, KPI analysis, trend analysis, and revenue analysis.

Tools & Technologies: Power BI, Power Query, DAX, and Excel.

5. Expected Outcomes

The project is expected to deliver an interactive Hospital Management Dashboard with insights into patient demographics, appointment trends, doctor performance, treatment costs, and revenue reports. The dashboard will help improve operational efficiency and healthcare decision-making.

6. Risks and Challenges

Missing or inconsistent data may require additional cleaning. Incorrect relationships may affect dashboard accuracy. Large datasets may impact dashboard performance and visualization readability.

7. Conclusion

This project demonstrates how Power BI and data analytics can improve hospital operations and healthcare management. The Hospital Management Analytics Dashboard will provide meaningful insights for better operational transparency and data-driven decision-making.
